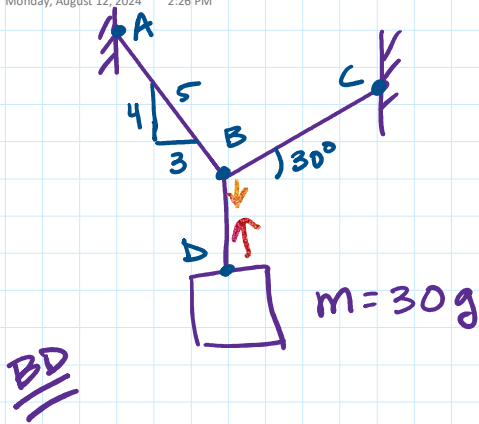
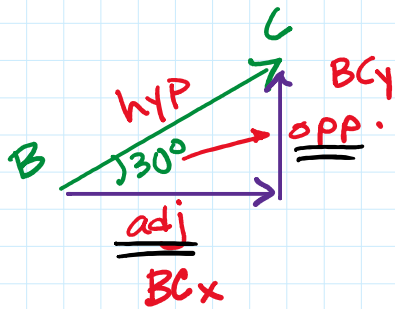




* Equilibrium
 $a = 0$

* $\sum \vec{F} = 0$
↑
sigma (sum)



→
BC

Soh Ca hToa

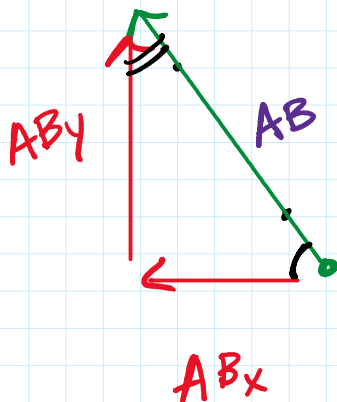
$$(BC) \sin \theta = \frac{BCy}{BC} (BC)$$

$$BCy = BC \cdot \sin \theta$$

↓ longest side ↓ < 1

$$BCy = BC \cdot \sin \theta$$

$$BCx = BC \cdot \cos \theta$$



* The Δ 's are similar

* Corresponding sides are proportional

* Scale one triangle to get the other one.

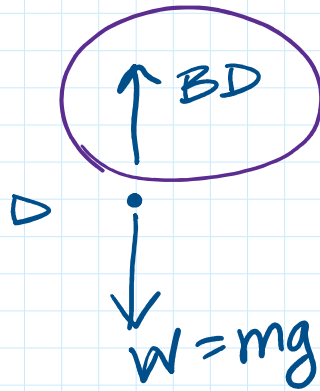
$$ABx = \frac{3}{5} AB = 0.6 AB$$

$$ABy = \frac{4}{5} AB = 0.8 AB$$



3

FBD



$$\sum F_y = 0$$

$$BD - W = 0$$

$$BD = W$$

$$BD = \underline{mg}$$

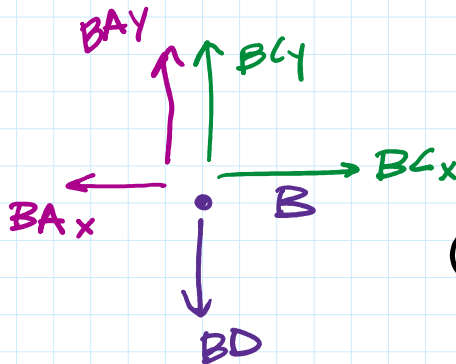
$$BD = (0.03)(9.81)$$

$$BD = 0.29 \text{ N}$$

$$1 \text{ Newton} = 1 \text{ kg} \cdot \text{m/s}^2$$

$$\frac{30 \text{ g}}{1} \left(\frac{1 \text{ kg}}{1000 \text{ g}} \right) = 0.03 \text{ kg}$$

FBD



$$\sum F_x = 0$$

$$BC_x - BA_x = 0$$

$$BC_x = BA_x$$

①

$$BC \cos \theta = 0.6 BA$$

$$\sum F_y = 0$$

$$BA_y + BC_y - BD = 0$$

②

$$0.8 BA + BC \cdot \sin \theta = BD$$

1a

$$BA = \frac{BC \cos(30^\circ)}{0.6} = 1.44 BC$$

$\Rightarrow$

$$BA = 1.44 BC$$

2a

$$0.8 (1.44 BC) + BC \cdot \sin(30^\circ) = 0.29$$

(2a)

$$0.8(1.44 BC) + BC \cdot \sin(30^\circ) = 0.29$$

$$1.15 BC + 0.5 BC = 0.29$$

$$1.65 BC = 0.29$$

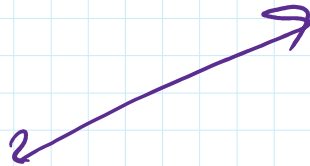
$$BC = 0.175 \text{ N}$$

$$BA = 1.44(0.175 \text{ N})$$

$$BA = 0.25 \text{ N}$$

Brendon

$$BA = 0.26 \text{ N}$$



Right answer

$$\frac{\Delta}{\text{Right}} = \frac{0.01}{0.25} = 0.04 \times 100\% = 4\%$$